Saibu Baba

Using Yelp Data & Data-Driven Strategies to Guide CoffeeKing’s Market Entry Strategy in Nashville, TN.

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- Approach: selecting metrics and dataset to answer questions, and provide insights for CoffeeKing

PART 1

- This project uses a filtered subset of the Yelp Academic Dataset focused on Nashville to support CoffeeKing’s market entry strategy.

- In this phase, the dataset will be imported into a SQL environment (Databricks) and cleaned by handling missing values, standardizing formats, and removing duplicates.

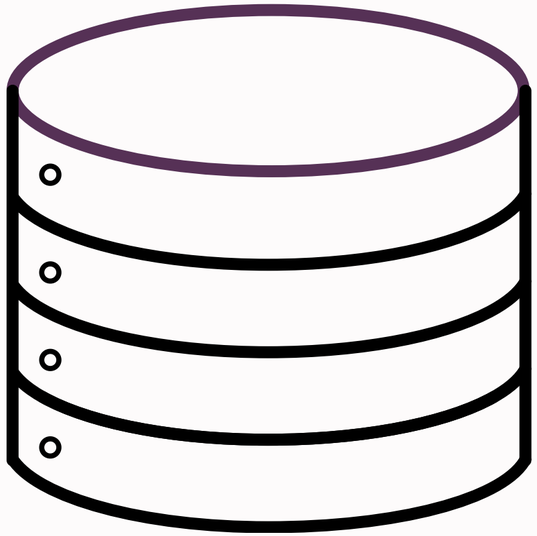
- An initial exploratory analysis will then be conducted to understand key variables, followed by proposing a star-schema ER diagram to clearly define relationships between the core tables.

Choosing the dataset

The Dataset

Selected dataset

For this project, I went with the CoffeeKing Dataset from Yelp. As a data analyst, I will use data-driven strategies to help th startup launch in Nashville.



Problem Statement

CoffeeKing is a new startup Coffee company providing unique and novel experience to their customers in Nashville, TN.

They want to appeal to a wide variety of clientele.

You will use Yelp data to provide insights to CoffeeKing for things such as location selection and or hours of operation or any other questions you want to explore based on the provided data

Why this project & Dataset

Why this project and dataset

```
Company Name: CoffeeKing
Potential location: Nashville, TN
```

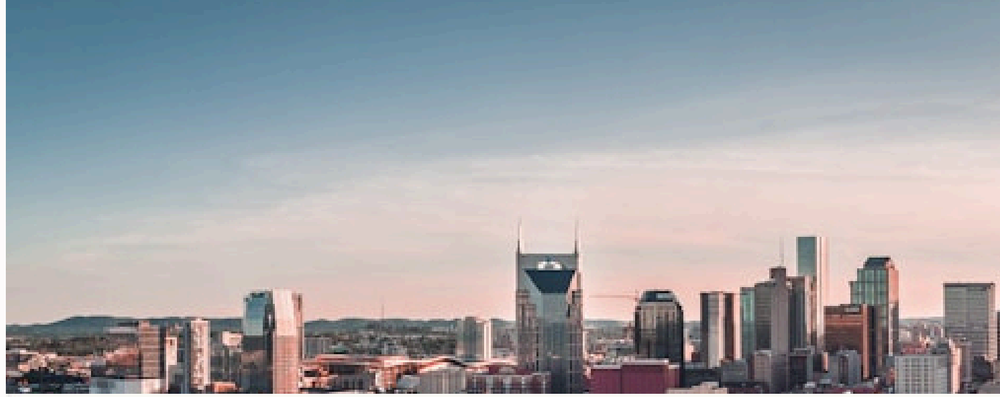
Goal of CoffeeKing
They want to appeal to a wide variety of clientele.

My goal as a data analyst.

- To help CoffeeKing choose the best location for its operations in Nashville.
- To help the startup select the most optimal times to drive more sales and increase demand for its coffee
- To back CoffeeKing's launch in Nashville with data-driven strategies

Understanding the location: Nashville

For this section, I wanted to understand the city a bit. Since this is external data from Yelp, a small overview of the city could help answer questions contextually and tailor insights properly to CoffeeKing



Key locations worth noting

1. Down town/Broadway - heart of NV nightlife and live music.
2. The Gulch upscale urban area, luxury hotels, restaurants, shopping
3. East Nashville - Trendy, creative, and vintage vibe
4. 12South - highly walkable neighborhood, boutiques & dining
5. Grand Ole Opry
6. The Parthenon - a replica of original in Athens, Centennial Park

Demographic Info.

Population: 705,000
Metro Area: 1.2 Million
State: Tennessee

Educ. Music, & Culture

- Popular sch: Vanderbilt Uni
- Nick name: Music City or Country Music Capital
- Country Music Hall of Fame & Museum

Importing and cleaning the dataset

1. I started by downloading the dataset from the Coursera course, where it was provided (the entire zip file).
2. I proceeded to extract the data, in order to get all the tables (5 in total) on business, user, review, tip, and checkin
3. Then I imported the tables one after the other into my workspace. I used Databricks for this project (it was a chance to also learn about how the platform works).

Initial exploration to understand the dataset

For this part, I performed a few queries to understand the data. That is, using DESCRIBE to check all the columns I have in the data. I also used COUNT to get the number of rows in each table.

Using DESCRIBE to find column info of all tables

Just now (2s) 1

```
%sql
DESCRIBE yelp_academic_dataset_nash_tip;
```

See performance (1) Optimi

_sqldf: pyspark.sql.connect.dataframe.DataFrame = [col_name: string, data_type: string

Table +

	col_name	data_type	comment
1	business_id	string	null
2	compliment_count	bigint	null
3	date	timestamp	null
4	text	string	null
5	user_id	string	null
6	_rescued_data	string	null

6 rows | 1.74s runtime

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SQL Editor
Queries
Dashboards
Genie
Alerts
Query History
SQL Warehouses

Just now (2s) 1 SQL

```
%sql
DESCRIBE yelp_academic_dataset_nash_business;
```

See performance (1) Optimize

_sqldf: pyspark.sql.connect.dataframe.DataFrame = [col_name: string, data_type: string ... 1 more field]

Table +

	col_name	data_type	comment
1	address	string	null
2	attributes	string	null
3	business_id	string	null
4	categories	string	null
5	city	string	null
6	hours	string	null
7	is_open	bigint	null
8	latitude	double	null
9	longitude	double	null
10	name	string	null
11	postal_code	bigint	null
12	review_count	bigint	null
13	stars	double	null
14	state	string	null
15	_rescued_data	string	null

15 rows | 1.82s runtime Refreshed now

This result is stored as `_sqldf` and can be used in other Python and SQL cells. Send feedback

I used COUNT too to count the number of rows in each of the tablese

Just now (3s) 1

```
%sql
SELECT COUNT(*) FROM yelp_academic_dataset_nash_business
```

See performance (1)

_sqldf: pyspark.sql.connect.dataframe.DataFrame = [COUNT(*): long]

Table +

	COUNT(*)
1	6971

Just now (3s) 1

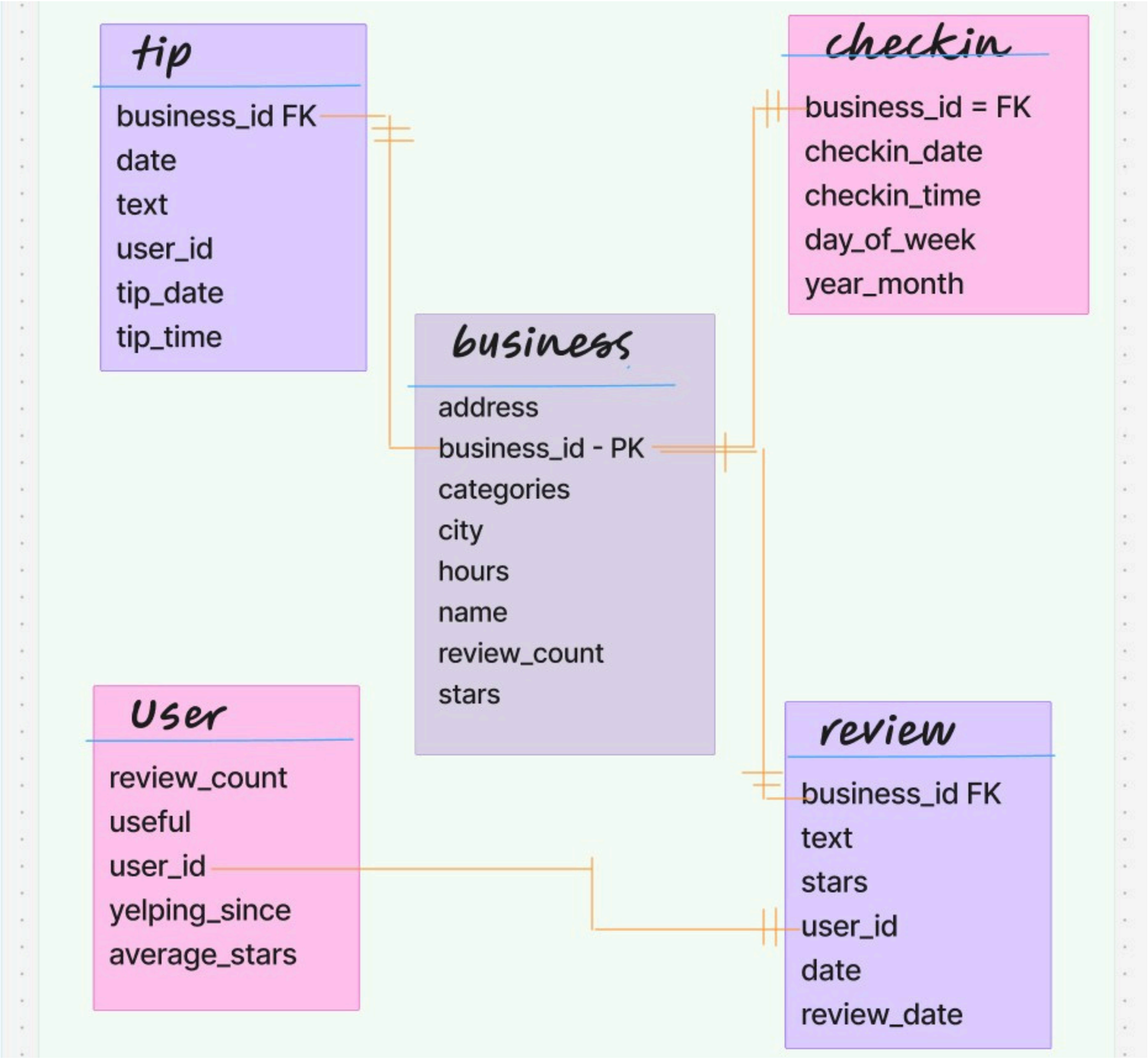
```
%sql
SELECT COUNT(*) FROM yelp_academic_dataset_nash_user
```

_sqldf: pyspark.sql.connect.dataframe.DataFrame = [COUNT(*): long]

Table +

	COUNT(*)
1	1166966

Proposed ER Diagram



PART 2

- This part, (as specificed by the course), will focus the project description and what I hope to achieve at the end of the analysis and insights.

- This second part will also cover the questions that I intend to answer as a data analyst on this project and my approach.

- Finally, we will explore my hypothesis and the data needed to prove or disprove those hypothesis.

Project Description

Analyzing Yelp Data

This project seeks to analyze YELP Data from Nashville, to uncover insights that will help CoffeKing make smarter decisions about entering the local coffee market.

The focus area

The focus of the analysis will be on finding the best locations in Nashville for CoffeeKing's operations. We'll also explore the best hours for CoffeeKing to leverage if they want to sell fast, and understand customer preferences. We'll do this by exploring ratings and business patterns

Our audience

The insights will guide CoffeeKing's strategy and improving its chances of success. Main audience includes CoffeeKing's founders and management team. Other interested audiences could include investors, business analysts, and anyone studying market entry strategies.

Questions to answer for CoffeeKing

In order to drive insights and give CoffeeKing the best advice on entering the Nashville market, here are 3 important questions worth exploring.

1. Location

Where should CoffeeKing locate its business if it wants to get more visibility and drive sales?

2. Optimal times/hours

What times should CoffeeKing operate to capture a chunk of the population (peak demand) and to maximize profit?

3. Customer experience

What's the state of customer experience for food and beverage brands in Nashville? What's working? What's not? (using ratings, reviews, etc).

Hypothesis

01

Central/Busy areas of Nashville would already have a great deal of coffee shops. They'll also have higher overall ratings and review because of the sheer number of customers.

02

Coffee shops that open earlier and close later will receive more reviews, suggesting they capture more customer traffic than those with limited hours.

03

Many coffee shops may not fully cover peak demand hours (early mornings or late evenings), creating an opportunity for CoffeeKing to differentiate.

Approach: prove/disprove hypothesis & answer questions for insights

Features to look at first:

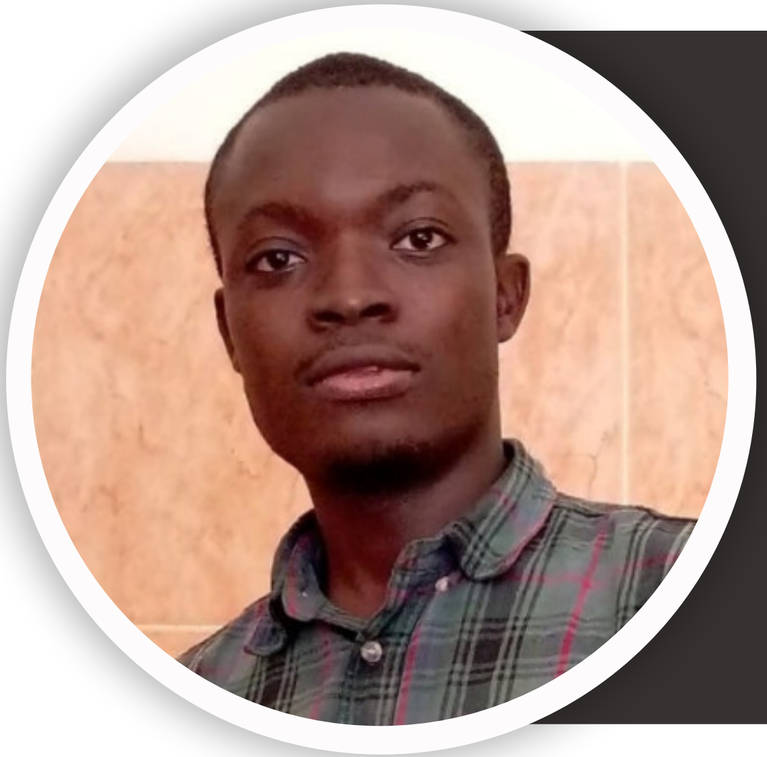
The analysis will begin with the business table, focusing on latitude, longitude (these two columns I removed them earlier because I didn't think through and thought they were not of use), stars, review_count, and hours to understand where Nashville coffee shops are located, how they are rated, and when they operate. The checkin_date, hour_of_day, and day_of_week columns from the cleaned check-in table will also be examined early to understand when customer activity actually peaks.

Relationships to explore

- Location vs. ratings/reviews. Eg, do coffee shops in central, busier areas of Nashville have higher stars and review_count than those in quieter neighborhoods?
- Operating hours vs. review count. Is there a relationship between how long a coffee shop stays open and how many reviews it accumulates?
- Peak demand hours vs. actual opening hours: Where are the gaps between when customers are most active and when shops are actually open?

Metrics/evaluation measures

- Average stars and total review count — to measure popularity and customer satisfaction by location
- Pearson correlation — to test whether longer operating hours predict higher review counts
- Frequency counts and hour-of-day distributions — to identify underserved time slots in early mornings and late evenings that represent an opportunity for CoffeeKing



THANK TOU

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